CAP 6908 DIS (Fall 2026)

When does a LiDAR detector stop relying on the object? A stratified study of evidence dependence in 3D detection confidence.

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Week of Aug 31 - Literature review of attribution methods for 3D detectors and of range and density effects on detector confidence. Deliverable: annotated bibliography of the reading list, plus a one-page problem statement fixing the exact claim to be tested and positioning it against the OccAM's Laser point-dropping result.

Progress: I read the full 14-paper syllabus list this week and wrote both deliverables: a 14-entry annotated bibliography and a one-page problem statement. Both are attached below in this single PDF. These are the week-2 assignment deliverables; the final report later in the semester will be in CVPR paper format, as discussed.

The bibliography covers all 14 papers. OccAM's Laser is the prior result the claim is written against. PointPillars is covered enough to state what the network sees (pillars, then a BEV pseudo-image) and that we are using the public OpenPCDet checkpoint, not retraining. KITTI is skimmed for val versus test and easy / moderate / hard. The density, range, and calibration cluster (PDV, GACE, Calib3D, Guo) is cited as known behavior, not as methods we will implement. FFAM is there so OccAM is not the only attribution paper. SECOND and PV-RCNN are thin notes marked as week-9 comparisons. The four context papers (frustum attack, CAM attack, density-aware thresholding, Robo3D) are one paragraph each.

The problem statement locks the claim in one paragraph: hold the public PointPillars checkpoint fixed, measure how much confidence depends on points inside the predicted box versus the surrounding scene, and ask whether that dependence is stronger when the object is far, sparse, or small. OccAM already showed residual confidence after emptying the box. The contribution is the stratified measurement, not a new detector or attribution method. The page also states the working locks (PointPillars, KITTI val, Car, AP@R11 as the sanity metric, reproduced 77.29 against the published 77.28, AP@R40 logged but not compared), names the three interventions (interior removal, empty-region placebo, context removal), and states what we will not do (no training, no new attribution method, no test-server submit, no detector survey). Pedestrian and Cyclist are a week-11 extension if the Car run is solid; one two-stage comparison is week 9.

Findings: OccAM's Laser (Schinagl et al., CVPR 2022) already established two things this study will not claim as new. First, when they dropped all points inside predicted boxes, both IoU and confidence stayed above zero; surrounding structure still supported the score. Second, a single voxel's contribution grows with range, which is why they corrected for it with density-aware sampling. What they did not measure is whether that leftover, context-backed confidence is concentrated on far, sparse, or small objects, with a placebo and a context-removal control, on one frozen public checkpoint. That is the gap.

Next Steps: Week of Sep 7 is the per-detection metadata pipeline. For every prediction on KITTI val, I will record range from the sensor, number of LiDAR points inside the predicted box, box dimensions, matched ground truth, and IoU. Deliverable: per-detection table over the full val split.

Attached Below:

1. Problem Statement (one page)
2. Annotated Bibliography (14 papers)

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Problem Statement

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Hold one public KITTI LiDAR checkpoint fixed. Measure how much its confidence depends on points inside the predicted box versus the surrounding scene, and whether that dependence is stronger when the object's own evidence is weak (far, sparse, or small). OccAM already showed residual confidence after emptying the box. The contribution is the stratified measurement, not a new detector or a new attribution method.

OccAM's Laser (Schinagl et al., CVPR 2022) is the prior result this study is written against. They built occlusion-based attribution maps by hiding voxels and watching the detection change. Two findings are already on the table. First, when they dropped points inside predicted boxes, both IoU and confidence stayed above zero after every interior point was gone. Surrounding structure still supported the score. Second, a single voxel's contribution grew with range, because near objects are densely sampled and far objects are not. They corrected that with density-aware sampling so maps could be compared across distance. That paper asks which points matter for an explanation map. It does not ask whether leftover, context-backed confidence is concentrated on far, sparse, or small objects.

That’s the gap. Confidence on distant and thinly sampled objects is already known to be poorly calibrated (PDV, GACE, Calib3D, Guo et al.). Those papers change the detector, the score, or the calibration. They do not hold one public checkpoint still and measure, by stratum, how much of the score comes from the box versus the scene. Other 3D attribution work exists (for example FFAM). None of it is this measurement.

What we add is the breakdown plus two controls. Keep the public OpenPCDet PointPillars checkpoint frozen. PointPillars sees the cloud as pillars, then as a bird's-eye-view pseudo-image; we are not retraining it. On KITTI val, Car only, record range, interior point count, and box size for each detection. Then apply three interventions: interior removal of points inside the predicted box, an empty-region placebo that removes a matched volume where there is no object, and context removal of points outside the box. Re-run the same detector and ask how much of the original confidence remains, and whether that retained fraction is larger when the object itself is weak. Exact removal code is a later methods step. This page only locks the claim and the three interventions.

Working locks: PointPillars is primary. Evaluation is KITTI val, never the test server. Class is Car. The sanity metric is official 3D AP@R11, Car moderate. Week 1 already reproduced 77.29 against the published OpenPCDet number 77.28. AP@R40 is logged so the wrong column is not compared to 77.28. KITTI easy / moderate / hard stay as published difficulty bins.

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Annotated Bibliography

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Benchmark and detectors

1. Geiger, Lenz, and Urtasun. Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite. CVPR 2012.

Geiger et al. introduced the KITTI suite from a station wagon instrumented with stereo cameras, a Velodyne HDL-64E, and a GPS/IMU unit. For objects they labeled 3D boxes in the laser cloud and projected them into the image, covering cars, vans, trucks, pedestrians, cyclists, and other classes, and they stood up an online evaluation server. The 2012 paper itself reports a smaller object experiment (about 12,000 images, 40,000 objects) and scores non-truncated, weakly occluded objects above a height cutoff at 50 percent box overlap. The later official 3D object benchmark that OpenPCDet implements is the protocol this study actually uses: 7,481 training frames, commonly split into train and val following Chen et al., and 7,518 test frames that only the server scores. Easy, moderate, and hard are the official difficulty bins (box height, occlusion, truncation). We evaluate on val. We never submit to the test server. This paper does not tell us how a modern detector's confidence splits between interior points and the scene.

2. Lang et al. PointPillars: Fast Encoders for Object Detection from Point Clouds. CVPR 2019.

PointPillars is the primary detector and the only checkpoint this study runs this semester. Lang et al. bin the cloud into vertical columns on the ground plane, run a small PointNet in each pillar, scatter those features into a bird's-eye-view pseudo-image, and detect with a 2D CNN and an SSD head. There is no vertical voxel bin to tune. They trained on lidar only and reported KITTI test numbers for car, pedestrian, and cyclist in both BEV and 3D, with a 62 Hz setting and a faster 105 Hz setting. We are not retraining that network. We use the public OpenPCDet zoo weights. Week 1 already reproduced Car moderate 3D AP@R11 of 77.29 against the published 77.28; AP@R40 is logged and is not compared to 77.28. This paper tells us what the network sees (pillars, then BEV). It does not measure how much of a frozen score comes from points inside the box versus context, or whether that split changes with range, interior count, or size.

3. Yan, Mao, and Li. SECOND: Sparsely Embedded Convolutional Detection. Sensors, 2018.

SECOND is a one-stage voxel detector. Yan et al. keep a VoxelNet-style grouping of points into voxels, then replace dense 3D convolution with sparse convolution so the z-axis can be processed without paying for empty space. They add an angle-loss fix for the pi flip in yaw and a ground-truth sampling augmentation. They report KITTI 3D results and a small real-time model at about 0.025 s on a GTX 1080 Ti. This is a later-comparison architecture, not this week's detector. Week 9 is when a second architecture matters. This paper does not answer the

stratified interior-versus-context question, and we are not implementing sparse convolution or their augmentation.

4. Shi et al. PV-RCNN: Point-Voxel Feature Set Abstraction for 3D Object Detection. CVPR 2020.

PV-RCNN is a two-stage detector that runs a 3D voxel CNN for proposals, summarizes the scene into keypoints, then uses PointNet-style set abstraction and RoI-grid pooling to refine boxes and confidences. Shi et al. evaluate on KITTI and Waymo and released the method in OpenPCDet. This is the planned week-9 two-stage check, used only to ask whether an evidence-dependence pattern is PointPillars-specific. It is not a second primary detector this week. The paper does not stratify residual confidence after interior removal.

Attribution and explanation

5. Schinagl, Krispel, Possegger, Roth, and Bischof. OccAM's Laser: Occlusion-Based Attribution Maps for 3D Object Detectors on LiDAR Data. CVPR 2022.

This is the paper the claim is written against. OccAM builds a point-level attribution map for each detection by randomly hiding voxels and watching the black-box output change. They do not need weights, activations, or gradients, so the same procedure runs on PointPillars, SECOND, and PV-RCNN. Because LiDAR density falls with range, they set each voxel's keep-probability from a fit to occupancy versus distance; a fixed keep-probability only made maps that were expressive in a narrow range band. They then drop points inside predicted boxes in most-important or least-important order and track confidence and IoU against the original detection. Two results are already known and are not our contribution: IoU and confidence stay above zero after every interior point is gone (their Figure 7 still has PointPillars detections on empty boxes because of ground cutouts), and a single voxel's contribution grows with range unless sampling is density-aware. What OccAM does not answer is whether that leftover, context-backed confidence is concentrated on far, sparse, or small objects, measured on one frozen public checkpoint with an empty-region placebo and a context-removal control. We are not building a new attribution method.

6. Liu, Li, Fang, Cui, and Huang. FFAM: Feature Factorization Activation Map for Explanation of 3D Detectors. NeurIPS 2024.

Liu et al. factor 3D backbone feature maps with non-negative matrix factorization, aggregate those concept maps into a global explanation, then refine the map with the feature gradient of a chosen object. They add voxel upsampling because linear interpolation does not fit a sparse cloud. They criticize OccAM as slow and sensitive to how many random subsets you draw, because each subset is a full forward pass. They report qualitative and quantitative explanation tests on several detectors and datasets. FFAM needs internals (3D features and gradients). OccAM does not. We will not implement FFAM. It does not hold a public checkpoint still and

report confidence retention after interior removal, placebo removal, and context removal, broken out by range, interior point count, and size.

7. You, Xu, and Demetriou. Explainability-Aware Frustum Attack: Exposing Structural Vulnerabilities in LiDAR-Based 3D Object Detectors. arXiv:2606.29963, 2026.

You et al. aggregate Integrated Gradients across scenes into a universal LiDAR saliency map (SALL), then attack only the most salient frustums (EFA) instead of whole object regions. On KITTI and nuScenes, with detectors including PointPillars and SECOND, they report more than a 15 point drop in recall while perturbing 25 to 50 percent fewer frustums than a non-saliency baseline. The useful fact for us is that evidence is concentrated in a small set of spatial regions. This is an attack paper. It is context for the claim, not the claim. We are not attacking the detector and we are not using their saliency maps as our metric.

8. Cai, Wang, Sohel, and Lei. Contextual Attribution Maps-Guided Transferable Adversarial Attack for 3D Object Detection. Remote Sensing, 2024.

Cai et al. build attribution maps over context subregions, then add perturbations there so the object itself stays visually intact. They report, in their abstract, an attack success rate above 68 percent on three large-scale sets and 83 percent on KITTI, and at least 50 percent transfer across four victim detectors, which they read as those detectors relying too much on context. That is consistent with OccAM's leftover-confidence observation, but it is still an attack result. We are not implementing CAMGA, and a success rate under perturbation is not a stratified retention table on a frozen zoo checkpoint.

Point density, range, and confidence reliability

9. Hu, Kuai, and Waslander. Point Density-Aware Voxels for LiDAR 3D Object Detection. CVPR 2022.

PDV is a two-stage detector built around the fact that LiDAR points diverge with range, so far objects return fewer points. Hu et al. localize voxel features at point centroids, pool with kernel density estimation and a density positional encoding, and then feed box centroid plus log interior point count into the confidence head. They report strong Waymo numbers and competitive KITTI val and test AP, using the official easy / moderate / hard bins and the Chen et al. train/val split. The known behavior we need is that density is a function of range and that confidence can be made to depend on it on purpose. We are not implementing PDV, KDE pooling, or their confidence head. They change the detector. They do not measure how a frozen PointPillars score splits between interior and scene after point removal.

10. Schinagl, Krispel, Fruhwirth-Reisinger, Possegger, and Bischof. GACE: Geometry Aware Confidence Enhancement for Black-Box 3D Object Detectors on LiDAR-Data. ICCV 2023.

GACE is from the same group as OccAM. They leave the detector frozen as a black box and learn a cheap extra head that rescores each box from geometry and neighbor context, because the usual classification head never sees heading, size, or local sparsity. They show that sparse or atypical sampling (their example is an occluded pedestrian) produces bad confidence, and that an oracle TP/FP sort of SECOND on Waymo would be worth several AP, especially on pedestrians and cyclists. They report gains on Waymo and also run KITTI, including PointPillars. Known behavior: raw detector confidence is a weak match to correctness when the object is thinly sampled. We are not adding a GACE rescoring head. They improve the score. They do not stratify residual confidence after emptying the box.

11. Kong et al. Calib3D: Calibrating Model Preferences for Reliable 3D Scene Understanding. WACV 2025.

Calib3D is a reliability benchmark: 28 models, 10 datasets, scored with expected calibration error. Kong et al. find that strong 3D models are often still badly calibrated. Uncalibrated models are especially over-confident in the middle-to-far band around the ego vehicle, where accuracy is already worse. Their fix is DeptS, a depth-dependent temperature on the logits. Known behavior: confidence and correctness diverge with depth. We will report reliability diagrams in week 4 as the baseline, not as a new calibrator. We are not implementing DeptS. Calib3D is mostly semantic scene understanding and uncertainty, not a counterfactual interior-versus-context study on KITTI Car detections.

12. Lee, Jung, and Kim. Toward Robust LiDAR based 3D Object Detection via Density-Aware Adaptive Thresholding. ICRA Workshop on Field Robotics 2024.

Lee et al. replace a single confidence threshold with a threshold that changes with distance from the ego vehicle, because near objects are dense and high-confidence and far objects are sparse and low-confidence. They argue that a fixed cutoff creates false positives on nearby clutter (their example is bushes scored as vehicles) and misses far objects. We are not retuning thresholds as the contribution. A distance-dependent cutoff is not a measurement of how much of the score lives inside the box.

13. Guo, Pleiss, Sun, and Weinberger. On Calibration of Modern Neural Networks. ICML 2017.

Guo et al. studied calibration on image and document classifiers, where the confidence of the predicted class should match how often that class is right. They showed that then-modern nets (deep ResNets on CIFAR-100, and similar) were over-confident, that depth, width, weight decay, and BatchNorm all move calibration, and that temperature scaling, a one-parameter Platt variant, was a strong post-hoc fix. They measure calibration with expected calibration error and visualize it with reliability diagrams. This is not a LiDAR paper. We cite it as the definition of the reliability plots week 4 will draw, and as the reason "the detector said 0.9" is not the same as "it is right 90 percent of the time." We are not running temperature scaling as the contribution, and this paper has no interior-versus-context result.

Robustness under degraded input

14. Kong et al. Robo3D: Towards Robust and Reliable 3D Perception against Corruptions. ICCV 2023.

Robo3D is a corruption benchmark for 3D detectors and segmentors. Kong et al. add eight corruptions (fog, wet ground, snow, motion blur, missing beams, crosstalk, incomplete echo, cross-sensor) at three severities and show that models that look strong on clean val sets drop under those shifts. They propose density-insensitive training and flexible voxelization as robustness knobs. This study does not add weather or sensor corruptions. Our interventions are interior removal, an empty-region placebo, and context removal on otherwise clean KITTI val clouds.